Chapter 4, Section 5

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1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). □

2. If X is a curve of genus $\geqslant 2$ over a field of characteristic 0, show that the group $\operatorname{Aut} X$ of automorphisms of X is finite.

Proof. $\boxed{X \text{ is hyperelliptic.}}$ Let $\pi : X \to \mathbb{P}^1$ be the unique g^1_2 of X. By the Riemann-Hurwitz formula, π is ramified at $2g+2$ points. If σ is an automorphism of X, then there exists $\tau \in \mathrm{PGL}(1)$ such that the following diagram commutes

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^1 & \xrightarrow{\tau} & \mathbb{P}^1 \end{array}.$$

This means $\mathrm{Aut}(X)$ permutes the branch points of π, so there is a group homorphism $\mathrm{Aut}(X) \to S_{2g+2}$. The latter is a finite group, so it remains to show it has finite kernel. If σ fixes every branch point, then τ is the identity, so σ is either the identity map, or the unique hyperelliptic involution (think in terms of function fields and the fact that the hyperellitpic involution is induced by the unique non-trivial element in $\mathrm{Gal}(K(X)/K(\mathbb{P}^1))$).

$\boxed{X \text{ is nonhyperelliptic.}}$ Consider the canonical embedding $X \hookrightarrow |K| \cong \mathbb{P}^{g-1}$. If σ is an automorphism of X, then there exists $\tau \in \mathrm{PGL}(g-1)$ such that we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^{g-1} & \xrightarrow{\tau} & \mathbb{P}^{g-1} \end{array}.$$

By (Ex. 4.6), X has $g(g^2-1)$ hyperoscullation points in this embedding, and τ must permute them because linear transformations preserve incidence structures. So there is a group homomorphism $\mathrm{Aut}(X) \to S_{g(g^2-1)}$, and since $S_{g(g^2-1)}$ is finite, it remains to show the kernel is finite. For every $g \geqslant 3$, $X \cap H$ can contain at most point two hyperoscullation points for any hyperplane $H \subset \mathbb{P}^{g-1}$, i.e., no three hyperoscullation points are collinear. It is a general fact that any automorphism in $\mathrm{PGL}(n)$ is identity if it fixes at least $n+2$ points in general position. Since $g(g^2-1) \geqslant g+1$ for every $g \geqslant 3$, if τ fixes the hyperoscullation points, it must be the identity. Hence, σ is the identity. As a corollary, any automorphism of a nonhyperelliptic is induced by an automorphism of the canonical embedding, i.e., there exists an injective map $\mathrm{Aut}(X) \to \mathrm{PGL}(g-1)$. □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g^1_3 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbb{P}^1$, ramified at $0, 1, \infty$, and $2g-1 = 7$ additional points, up to the action of a certain finite group. So the hyperelliptic curves form an irreducible subvariety of dimension 7 of $\mathfrak{M}_4$ (5.5.5).

The canonical model of a nonhyperelliptic curve of genus 4 is the complete intersection of a unique quadric surface and a cubic surface in $\mathbb{P}^3$ (5.2.2). The family of all these curves is parametrized by the space $\mathfrak{S}$ of all complete intersections of a quadric surface and a cubic surface. We can think of $\mathfrak{S}$ as the set of all homogenous ideals generated by a degree two and three homogenous polynomial in $k[x_1, \dots, x_4]$. So there is a morphism $\mathfrak{S} \to \mathfrak{M}_4$, whose fibres are images of the group $\mathrm{PGL}(3)$, which has dimension 15. It remains to compute the dimension of S. There is a projection map $\mathfrak{S} \to \mathfrak{Q}$, where $\mathfrak{Q}$ is the space of all irreducible quadric surfaces in $\mathbb{P}^3$. It is a locally closed subset of $H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2))|$, which has dimension $\binom{3+2}{2} - 1 = 9$. The fibres of $\mathfrak{S} \to \mathfrak{Q}$ is the image of pullback of the linear system of cubic surfaces in $\mathbb{P}^3$, which is just $|H^0(Q, \mathcal{O}_Q(3))|$ for some quadric surface $Q \subset \mathbb{P}^2$. Note that the dimension of this space is independent of the choice of Q. Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_Q \longrightarrow 0.$$

Tensoring by $\mathcal{O}_{\mathbb{P}^3}(3)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(1)) \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow H^0(Q, \mathcal{O}_Q(3)) \longrightarrow 0.$$

By exactness, we have

$$\dim |H^0(Q, \mathcal{O}_Q(3))| = \binom{3+3}{3} - \binom{3+1}{1} - 1 = 15,$$

which implies the fibres of $\mathfrak{S} \to \mathfrak{Q}$ have dimension 15, so $\mathfrak{S}$ has dimension $9+15 = 24$. Hence, $\mathfrak{M}_4$ has dimension $24 - 15 = 9$. The subspace of $\mathfrak{Q}$ consisting of quadric cones have dimension 8, so the space of these curves have codimension one in $\mathfrak{M}_4$. Hence, the subset of of nonhyperelliptic curves having only one g^1_3 is an irreducible family of dimension 8. □

5. *Curves of Genus 5.* Assume X is not hyperelliptic.

(a) The curves of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection $F_2.F_2.F_2$ form a family of dimension 12.

(b) X has a g^1_3 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.

(c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbb{P}^2 \to \mathbb{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbb{P}^4$, with V isomorphic to $\mathbb{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g^1_3 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X. Thus V and g^1_3 are unique.

Proof.

(a) A curve of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection of three quadrics is given by a three-dimensional subspace of $H^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2))$, which has dimension $\binom{4+2}{2} = 15$. Thus, the family of all these curves is parametrized by an open subset of the Grassmannian $\mathrm{Gr}(3, 15)$, which has dimension $3(15-3) = 36$. There is a natural action of $\mathrm{PGL}(4)$ on $\mathrm{Gr}(3, 15)$, and since the embedding is canonical, two three-dimensional subspaces in $\mathrm{Gr}(3, 15)$ give isomorphic curves if and only if they are in the same orbit of $\mathrm{PGL}(4)$. So there is a morphism $\mathrm{Gr}(3, 15) \dashrightarrow \mathfrak{M}_5$, whose fibres are the images of the group $\mathrm{PGL}(4)$ which has dimension $(4+1)^2 - 1 = 24$. Since any individual curve has only finitely many automorphisms (Ex. 5.2), the fibres have dimension $= 24$, and so the image of $\mathrm{Gr}(3, 15)$ has dimension $36 - 24 = 12$.

(b) A g^1_3 on X induces an embedding of X in the nonsingular quadric surface Q as a divisor of type $(3, 3)$. Under the standard embedding of Q into $\mathbb{P}^3$ gives an embedding $X \to \mathbb{P}^3$ where the image is a degree $3 + 3 = 6$ curve in $\mathbb{P}^3$. Projecting X from one of its points that does not lie on a plane spanned by coplnar tangent lines gives a plane quintic with one node.
Conversely, suppose X is represented let $\pi : V \to \mathbb{P}^2$ be the blowing up of $\mathbb{P}^2$ at the node P of $X \to X' \subset \mathbb{P}^2$ as a plane quintic curve , and let $E \subset V$ be the exceptional divisor of π. Let H be any

hyperplane in $\mathbb{P}^2$ so that $|2H|$ is the linear system of conics in $\mathbb{P}^2$. Using the results of (V, §3) and (V, §4), the linear system $|2H-P|$ of conics passing through P has dimension $5-1=4$ with an assigned basepoint at P. By (II, 7.17.3), the rational map $\mathbb{P}^2 \dashrightarrow |2H-P| \cong \mathbb{P}^4$ extends to a morphism $V \to \mathbb{P}^4$. It is an embedding of V into $\mathbb{P}^4$ because it corresponds to the linear system $|D|$ with $D = 2\pi^*H - E$ on V, which is very ample by (V, 4.2.2). The strict transform $\tilde{X}'$ of X' is nonsingular, i.e., $\tilde{X}' \cong X$, and by (V, 3.6), it relates to X' as divisors in V in the following way

$$\pi^*X' = \tilde{X}' + 2E.$$

Using the fact that $X' \sim 5H$ in $\mathbb{P}^2$, by (V, Ex. 1) the degree of $\tilde{X}'$ as a curve in $\mathbb{P}^4$ equals to

$$\tilde{X}'.D = (5\pi^*H - 2E).(2\pi^*H - E) = 10(\pi^*H)^2 + 2E^2 = 10 - 2 = 8.$$

By the Riemann-Roch theorem, the only degree $2g-2$ invertible sheaf with at least g sections on a curve of genus g is the canonical sheaf, so this embedding of X in $\mathbb{P}^4$ is canonical. Now, the identification of V with a rational scroll $V \to \mathbb{P}^1$ of degree 3 in $\mathbb{P}^4$ (V, 2.19.1) induces the following correspondence of divisors on V

$$a\pi^*H - bE = (a-b)C_0 + af,$$

where C_0 is a section and f is any fibre of the ruling $V \to \mathbb{P}^1$. Rewriting $\tilde{X}'$ in this way gives

$$\tilde{X}' = 3C_0 + 5f.$$

Since $\tilde{X}'.f = 3$, a fibre of $V \to \mathbb{P}^1$ and $\tilde{X}'$ intersect at three points. Also, V is a rational scroll in $\mathbb{P}^4$, so any fibre of V is a line in $\mathbb{P}^4$. Hence, any fibre of $V \to \mathbb{P}^1$ is a trisecant of X, so X has a 1-parameter family of trisecants in $\mathbb{P}^4$.

By (3.11.1), the space $V_{d,r}$ of irreducible plane curves of degree d and r nodes is irreducible and has dimension $\frac{1}{2}d(d+3) - r$. In our case, $V_{5,1}$ has dimension 19, and we have just shown $V_{5,1}$ parametrizes the space of all genus 5 curves with a g^1_3. So there is a morphism $V_{5,1} \to \mathfrak{M}_5$, whose fibres are images of the group PGL(2), which has dimension 8. Hence, the family of genus 5 curves with a g^1_3 form an irreducible family of dimension $19 - 8 = 11$.

(c) The results of this exercise combined with (5.5.3) can be summarized by the following commutative diagram

$$\begin{tikzcd}
 & \mathbb{P}^2 & \\
X \arrow[ur, "\text{plane quintic}"] \arrow[r, hook] \arrow[dr, "g^1_3"'] & V \arrow[u, "\text{blow up}"'] \arrow[r, hook] \arrow[d, "\text{ruled surface}"] & \mathbb{P}^4 \cong |K| \cong |D| \\
 & \mathbb{P}^1 &
\end{tikzcd},$$

where K is a canonical divisor of X, and D is the divisor of V from (Ex. 5.5b). Thus, we have three different representations of a nonhyperelliptic curve of genus 5 with a g^1_3.

It would be intersection to know what the map of $X \hookrightarrow V$ would like if I know the g^1_3. In particular, defining a morphism $X \to V$ is equivalent to defining a surjective morphism of $\mathcal{O}_X$-modules $f^*\mathcal{E} \to \mathcal{L}$, where $\mathcal{E} = \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-1)$ and $\mathcal{L}$ is an invertible sheaf on X.

□

6. Show that a nonsingular plane curve of degree 5 has no g^1_3. Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

Proof. Let $X \subset \mathbb{P}^2$ be a nonsingular plane curve of degree 5. The Veronese embedding $\mathbb{P}^2 \xrightarrow{\sim} V \subset \mathbb{P}^5$ realizes $\mathbb{P}^2$ as a surface of degree 4, and realizes X as a curve of degree 10 in $\mathbb{P}^5$. Since $g = 6$, this embedding is the canonical embedding. In particular, the only degree $2g-2$ invertible sheaf with at least g sections on a curve of genus g is the canonical sheaf. The proof of (5.5.3) applies to any nonhyperelliptic curve. In particular, X has a g^1_3 if and only if its canonical embedding in $\mathbb{P}^5$ has a trisecant line. Suppose that $L \subset \mathbb{P}^5$ is a trisecant line to the canonical image of X, meeting X at three points P, Q, R. Since $X \subset V$, the line L meets the Veronese surface in P, Q, R.

These three points determine the g_3^1 on X, and hence $\dim|P+Q+R| = 1$. On the other hand, three points of the Veronese surface that are collinear in $\mathbb{P}^5$ must be collinear in $\mathbb{P}^2$, because the line through P, Q, R intersects V at three points, and V has degree 2 in $\mathbb{P}^5$, so the line must be contained in V. Thus P, Q, R are collinear in $\mathbb{P}^2$. But there is only one line in $\mathbb{P}^2$ passing through three distinct collinear points, contradicting the dimension of $|P+Q+R|$. Therefore, X has no trisecant line in its canonical embedding, and consequently X has no g_3^1.

For the second part, it is enough to show there exists nonhyperelliptic curves of genus 6 with a g_3^1. Consider any irreducible nonsingular curve X associated to a divisor of type $(3,4)$ on the nonsingular quadric surface Q. It has genus $(3-1)(4-1) = 6$, and one of the families of lines on Q cuts out a g_3^1 on X. Hence, X cannot be represented as a nonsingular plane quintic curve. The canonical sheaf of Q is $\omega_Q = \mathcal{O}_Q(-2,-2)$, so by (II, 8.20), the canonical sheaf of X is $\omega_X \cong \mathcal{O}_Q(1,2) \otimes \mathcal{O}_X$. Since $\mathcal{O}_Q(1,2)$ is a very ample, ω_X is very ample. Hence, X is nonhyperelliptic by (5.2). $\square$